

Project Report: Mortality Surveillance

Subtitle: A Retrospective Mortality Surveillance Analysis

Project Overview

The primary objective of this project report is to conduct a rigorous retrospective mortality surveillance analysis across 10 selected countries, focusing on identifying and ranking the absolute leading causes of death. By analyzing historical patterns of mortality by demographic distribution, the report isolates the apex threats within each cohort. This surveillance provides the precise, data-driven evidence required for targeted public health interventions, equipping policymakers with the strategic intelligence needed to optimize resource allocation, deploy aggressive countermeasures that will drastically reduce preventable deaths.

Problem Statement [Epidemiological Problem]

In this retrospective analysis, a historic baseline of existing records is evaluated to uncover systemic and environmental patterns of death. The core variables under investigation include Year, Country, Gender, Age Group, Cause of Death, Number of Deaths, and Mortality Rate (*per 1,000 people*).

This surveillance aims to answer four fundamental questions:

- **The Burden:** What are the documented baseline mortality rates, and what are their primary causes?
- **The Risk:** Which countries or populations experience the highest mortality rates?
- **The Disparity:** What is the demographic distribution of mortality rates by gender and age cohort?
- **The Trend:** What is the chronological trend of the average mortality rate over time?

By isolating the apex killer within each specific cohort and demographic, this report provides the empirical foundation necessary to aggressively redirect healthcare funding, personnel, and medical supplies to where they will save the most lives.

Objectives

The primary objective of this report is to analyze past mortality records across the 10 selected countries to help leaders make data-driven decisions. Specifically, this report aims to:

- Document overall mortality burden and isolate primary underlying causes of death.
- Identify which specific countries or population groups face the highest number of deaths.
- Stratify mortality data to evaluate differential trends across age cohorts and gender.
- Map out how these mortality shifts and patterns are changing overtime.

Ultimately, these objectives give policymakers the empirical evidence needed to direct healthcare financing, work force allocation and medical supply directly to high-priority areas, maximizing life-saving impact.

Data Source & Analytics

The dataset used for this analysis was sourced from Kaggle. Because the data was retrieved in a pre-cleaned, single-table format, it was immediately ready for import into Power BI. This allowed the focus to remain entirely on configuring key analytical measures (such as the average and total mortality rates), designing the interactive visualization, and conducting the epidemiological analysis.

The Core Data Fields

To get the right insights from this report, we need to understand how these columns translate into public health terms:"

- **Year (Time):** Allows us to look at *secular trends* (changes over time). It tells us if a disease is spiking or fading.
- **Country (Place):** Represents our 10 geographic cohorts across which mortality rates were compared.
- **Gender & Age in group (Person):** These are demographic risk factors. Diseases attack different age groups and sexes differently.
- **Cause of Death (Etiology):** The specific disease or reason for mortality.

- **No of Deaths (Absolute Burden):** The raw count of lives lost.
- **Mortality Rate (Relative Risk):** The number of deaths divided by the population size (usually expressed per 1000 people).

Methodology

To investigate systemic patterns of death and isolate the leading causes, the dataset was analyzed within Power BI using three sequential steps:

1. **Analyzing Total Deaths and Causes:** Overall death totals were grouped by specific causes for each country using bar charts and summary tables. This moved the analysis beyond raw numbers, exposing the unique impact of different health conditions and identifying which factor caused the highest volume of deaths in each nation.
2. **Identifying High-Risk Populations:** Power BI's interactive charts were used to compare total deaths across the 10 nations. This pinpointed which specific countries faced the heaviest overall burden of mortality.
3. **Analyzing Demographic Disparities:** Power BI's filtering tools were used to split the data by age group and biological sex. By comparing these groups in matrices and tables, the analysis exposed which specific life stages or genders are the most vulnerable.
4. **Tracking Trends Over Time:** Power BI's line charts were used to plot data chronologically by year. By analyzing these year-over-year patterns, the analysis monitored whether deaths from specific causes were rising, dropping, or remaining steady over time.

Data Visualization

To communicate these epidemiological insights effectively, an interactive Power BI dashboard was developed to translate raw mortality metrics into actionable visual intelligence for healthcare administrators. The design focuses on clean layouts and a quick visual scan.

Interactive Slicers: Universal filters for Year, Gender, Age Group, Country, and Cause of Death allow users to dynamically isolate specific sub-populations or health concerns across the entire reporting interface.

Geographic and Demographics Analysis: Stacked bar charts and matrix tables break down average mortality and total deaths simultaneously by Country, Gender, and Age

Group to quickly highlight the segments with vulnerable population (as shown on the 'Dashboard 1' tab). Line and clustered column charts give the granularity of total deaths and average deaths by country and by cause of death, while bar, donut, and column charts offer a direct view of average mortality across these demographic factors (on the 'Dashboard 2' tab).

Time Series Trend Analysis: A core feature of this surveillance report is the *Average Mortality by Year area chart*, which plots chronological data to monitor large-scale public health trends (on the Dashboard 1 tab)

To provide immediate clarity for data interpretation, a line, representing the average baseline was integrated directly into the area chart to serve as a reference point, instantly highlights key patterns:

Above-Average Peaks: Years where mortality rates spiked significantly above the baseline are immediately identifiable as they rise above the line, allowing administrators to cross reference with other dashboard charts to instantly isolate the leading causes of death behind the surge.

Below-Average Valleys: Years where mortality rates dropped below the baseline visually validate positive trends, allowing administrators to identify which public health interventions were effective.

To maximise data utility and provide a complete epidemiological view, this trend analysis responsively updates based on user selections. When administrators filter the data slicers, the area chart automatically recalculates, thus, allowing public health decision makers to tract trends and isolate specific epidemiological patterns within distinct population.

Key Findings & Surveillance Insights

This surveillance establishes an empirical baseline across ten global regions to optimize resource allocation and reduce preventable deaths. By analyzing death rates relative to population size rather than looking at raw counts alone, this report isolates true regional mortality risks and identifies the specific primary causes driving global deaths.

For the 10 regions, the dataset establishes an overall baseline benchmark of 6 deaths per 1,000 population across a population of 1.04 billion (accounting for 67 million total deaths). However, looking at individual countries reveals distinct country death rates and specific primary causes:

Baseline Mortality Rates and Leading Causes by Country

Country	Baseline Mortality Rate/1,000	Leading cause of death	Peak cause value
Nigeria	8	Cancer	11
Brazil	7	Suicide	10
Russia	7	Diabetes & Road Accidents	11
Pakistan	6	Stroke, Kidney Disease, & Homicide	9
Japan	6	Cancer	9
USA	6	Suicide	11
Germany	6	Cancer	9
China	5	Cancer	Aggregated
India	5	Respiratory Diseases	Aggregated
UK	5	Chronic Baseline Conditions	Aggregated

Core Surveillance Insights

Cancer turns to be the most dominant baseline threat, serving as the leading cause of death across four major population: Nigeria, Japan, Germany, and China.

While the average is 6, Nigeria (8), Brazil (7), and Russia (7) represent significantly higher overall mortality risks.

Non-disease factors also play a major role in death rates in some countries. This is distinctively clear in the USA and Brazil where Suicide leads and Russia where Road Accidents tie as a primary cause of death.

Countries experiencing the highest mortality rates

Based on the surveillance data, the risk of mortality is highly concentrated within specific geographic regions. The countries experiencing the highest overall baseline mortality rates are Nigeria, Brazil and Russia.

Nigeria carries the absolute highest mortality risk in the dataset with a baseline rate of 8. Brazil experiences an elevated mortality risk with a baseline rate of 7. Russia ties with Brazil, sustaining an elevated baseline mortality rate of 7.

These three regions systematically exceed the global average dashboard baseline of 6, representing the highest-risk countries under surveillance (Dashboard 2).

Demographic distribution of mortality rates by gender and age cohort.

Distribution by Gender

Based on the visuals on dashboard 2, the demographic distribution of baseline mortality rates indicates that Males carry a baseline average mortality rate of 7, accounting for the majority at 55.16% of the distribution while females carry a lower baseline average mortality rate of 5, representing 44.84% of the distribution.

An increased mortality rate among males can be attributed to the fact that males generally suffer significantly higher death rates from external causes like road accidents and homicide due to occupational hazards and risk-taking behaviors. Delayed medical intervention, leading to late-stage detection of critical conditions like cancer and heart diseases, consistent with World Health Organization's Global Health Estimate of 2000-2021 and Behavioral Risk Factor Surveillance Systems (BRFSS) is consistent with the male gender.

Distribution by Age Cohort

Looking at the Average Mortality by Age Group chart in (Dashboard 2), the baseline rates show a clear disparity across life stages, visually tracking across the chart

45–59 Group: Suffers the absolute highest average mortality rate, peaking near 7.

15–29 Group: Holds the second-highest position, tracking just above 6.

60+ Group: Follows closely in the third position, anchoring right at 6

0–14 Group: Registers a lower baseline rate, sitting at approximately 5.

30–44 Group: Documents the lowest baseline risk in the dataset, dropping down near 4.5.

Increase in mortality Peaks in the 45–59 Age Cohort is attributed to onset of Chronic Diseases. This window is the age when decades of cumulative health risks manifest into fatal clinical presentations like Stroke, Diabetes, and aggressive Cancers. It is also the age of peak Socioeconomic Stress in which high professional and family responsibilities during these years severely aggravate cardiovascular and behavioral health risks. (Standard WHO global mortality data)

The chronological trend of the average mortality rate over time (Dashboard 1).

Based on the surveillance data, the findings reveal a fluctuating, non-linear trend over the timeline rather than a constant trajectory.

In 2015, the average mortality rate experienced an upward trajectory, tracking distinctly above the baseline. Following the initial peak, the trend experienced a visible drop, showing a temporary reduction in the overall mortality rate. By 2018, there was

subsequent Resurgence followed by a downward trajectory and the average mortality rate rose again.

The overall trend is characterized by a fluctuating pattern, demonstrating that the mortality burden experienced periods of escalation, temporary stabilization, and subsequent resurgence rather than a constant decline or continuous increase.

Recommendations

1. Target Top Causes of Death:

Prevent External Injuries: Implement stricter road safety enforcement and occupational safety protocols to directly lower the high toll from Road Accidents and External Trauma.

Strengthen Chronic Disease Screening: Expand community access to early diagnostic screenings for Cancer and cardiovascular diseases to catch high-risk conditions before they become fatal.

2. Address Gender Disparities:

Improve Male Healthcare Access: Launch targeted health campaigns in male-dominated spaces and workplaces to encourage early medical checkups and break the habit of delaying care.

Focus Prevention on High-Risk External Factors: Direct injury-prevention and behavioral health resources specifically toward males, who face a skewed share of accidental and violent deaths.

3. Intervene by Age Cohort

Target the 45–59 Peak Risk Group: Prioritize this window for aggressive health screenings (blood pressure, blood sugar, oncology) to catch chronic diseases as they begin to surface.

Support Mental and Behavioral Health: Integrate robust mental health support into routine care for young and middle-aged demographics to address spikes in areas like suicide and stress-related risks.

Conclusion

This Mortality Surveillance project conducted a Retrospective Mortality Surveillance Analysis across 10 selected countries to identify, rank, and analyze the leading causes of death by geographic and demographic distribution. By evaluating historical records based on year, country, gender, age group, and mortality rates, the analysis established critical data-driven insights to help optimize public health resources and reduce preventable deaths. This surveillance revealed three definitive conclusions:

Risks by Gender: Mortality is heavily skewed toward males, driven significantly by higher exposure to external injuries and a documented tendency to delay seeking medical attention.

Vulnerability by Age: The 45–59 age cohort represents the highest risk window, where decades of cumulative chronic diseases and socioeconomic stressors peak.

Fluctuating Long-Term Trends: The chronological data shows an inconsistent, up-and-down trajectory—surpassing the baseline in 2015, dropping, and rising again in 2018—proving that mortality risks are systemic and require ongoing, tracking of death trends to guide public health interventions.

By supporting these public health insights in clear visual evidence, this analysis equips policymakers with the strategic intelligence needed to deploy targeted screenings, enforce injury prevention programs such as road safety and speed management, and proactively flatten global mortality curves.

Executive summary

This Mortality Surveillance project provides a Retrospective Mortality Surveillance Analysis across 10 selected countries to evaluate historical patterns of death and isolate apex threats across different demographic cohorts. Utilizing a dataset configured in Power BI, the analysis tracks key variables—including year, country, gender, age group, absolute death counts, and mortality rates—to provide public health leaders with the precise, data-driven evidence required to optimize resource allocation and reduce preventable deaths.

Mortality is heavily skewed toward males due to high exposure to external injuries and delayed medical care, with the 45–59 age cohort representing the highest risk window due to chronic disease accumulation. Chronological tracking reveals that these mortality risks are highly dynamic, showing an inconsistent, fluctuating trajectory that spiked in 2015 and rose again in 2026.

This retrospective analysis successfully translates raw mortality data into actionable visual intelligence via Power BI. To maximize life-saving impact, policymakers must move away from reactive models and implement ongoing, flexible tracking of death trends to dynamically deploy injury prevention programs, targeted screenings, and healthcare resources exactly where mortality risks surge.

Methodology to dashboard mapping (Just for my clarification)

Here is how your documented **Methodology** steps directly connect and map to the specific visual elements on your **Power BI dashboards** (Dashboard 1 and Dashboard 2):

1. Analyzing Total Deaths and Causes

Methodology: *"Overall death totals were grouped by specific causes for each country using bar charts and summary tables..."*

This step is directly represented by the foundational charts and high-level summaries that isolate what is killing people and where:

- **KPI Cards (Header on both dashboards):** The **Total deaths (67M)** and **Average Mortality/1000 (6)** cards establish the absolute high-level baseline of the overall burden.
- **Total deaths and Average Mortality by Cause of Death (Dashboard 2 - Top Left):** A combo bar and line chart that groups absolute death volume (bars) alongside average mortality rates (line) by specific causes (e.g., Road Accidents, Cardiovascular Disease).
- **Average Mortality by Country and Cause of Death (Dashboard 1 - Top & Bottom Left):** A horizontal bar chart identifying top combinations (e.g., Nigeria Cancer, Russia

Road Accidents) and a stacked 100% bar chart mapping out how causes of death are distributed proportionally across the countries.

2. Identifying High-Risk Populations

Methodology: *"Power BI's interactive charts were used to compare total deaths across the [countries]. This pinpointed which specific countries faced the heaviest overall burden..."*

This step is addressed by the geographical and macro-level comparative visualizations:

- **Total deaths and Average Mortality by Country (Dashboard 2 - Top Right):** A combo chart that visually ranks the countries by absolute volume (e.g., USA, UK, Pakistan) while comparing them to their corresponding mortality rates.
- **Summary Table (Dashboard 1 - Bottom Right):** A structured table listing each country alongside its **Total Population** and **Total deaths** (e.g., USA at 8.03M deaths, UK at 7.84M deaths), offering a clear, raw-number comparison of geographic burden.
- **Average Mortality by Country (Dashboard 2 - Bottom Left):** A clean horizontal bar chart ranking the countries by their average mortality index (with Nigeria leading at an index of 8).

3. Analyzing Demographic Disparities

Methodology: *"Power BI's filtering tools were used to split the data by age group and biological sex... comparing these groups in matrices and tables..."*

This step corresponds to your demographic stratification charts, which break down vulnerability by life stages and gender:

- **Total Deaths by Cause of Death and Gender (Dashboard 1 - Top Middle):** A clustered horizontal bar chart comparing female vs. male deaths across different causes (e.g., highlighting that male road accidents significantly outnumber female road accidents).
- **Average Mortality by Gender (Dashboard 2 - Bottom Middle):** A donut chart highlighting the overall percentage breakdown of mortality by sex (Male at 55.16% vs. Female at 44.84%).
- **Total Deaths by Cause of Death and Age Group (Dashboard 1 - Top Right):** A stacked bar chart segmenting causes of death across specific age cohorts (0-14, 15-29, 30-44, 45-59, 60+).
- **Average Mortality by Age Group (Dashboard 2 - Bottom Right):** A column chart ranking average mortality indexes across life stages, showing that the 45-59 age group holds the highest average index (7).
- **Slicers (Top Ribbon on both dashboards):** The interactive **Gender**, **Age_Group**, and **Country** slicers serve as the functional engines that allow policymakers to drill down and filter any visual on the page by demographic subsets.

4. Tracking Trends Over Time

Methodology: *"Power BI's line charts were used to plot data chronologically by year... to monitor whether deaths... were rising, dropping, or remaining steady..."*

This step is represented by your chronological tracking visualization:

- **Average Mortality by Year (Dashboard 1 - Bottom Middle):** A dynamic line chart plotting the average mortality rate year-by-year from 2015 to 2025. This visual highlights a dip in 2018 (index of 4), peaks in 2015, 2019, and 2022 (index of 7), and projects a steadying trend toward 2025.
- **Year Slicer (Top Ribbon):** Allows users to isolate specific multi-year periods to see if interventions implemented in a given timeframe successfully reduced mortality rates.